INCH-POUND

MIL-E-24763A(SH)
25 August 1993

SUPERSEDING
MIL-E-24763(SH)
19 June 1991

MILITARY SPECIFICATION

ENAMEL, EMULSION TYPE, FOR SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of [95 percent acrylic] water based emulsion (latex), enamels primarily for use on primed metal surfaces. The enamel shall be volatile organic content compliant (VOC), and shall be supplied in a variety of colors (see 6.2 and 6.5) for shipboard use.

1.2 Classification. Enamel covered by this specification is of the following types and classes, as specified (see 6.2):

1.2.1 Types. The types of enamel shall be designated by "Type I" or "Type II" as follows:

Type I - Air-drying, acrylic emulsion type with a VOC not greater than 340 grams per liter (g/L) (2.8 pounds per gallon [LB/GAL]).

Type II - Air-drying, acrylic emulsion type with a VOC not greater than 250 g/L. (2.1 LB/GAL).

1.2.2 Classes. The class of the enamel shall be one of the following:

Class 1 - High gloss, 85 percent minimum.

Class 2 - Medium gloss, 45 to 60 percent.

Class 3 - Low gloss, 15 to 30 percent.

Class 4 - Flat gloss, 5 to 15 percent.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03Q42, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8010

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-A-250/4	- Aluminum Alloy 2024, Plate and Sheet
TT-P-645	- Primer, Paint, Zinc-Molybdate, Alkyd Type
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Sheets.
PPP-P-1892	- Paint, Varnish Lacquer, and Related Materials; Packaging, Packing and Marking of.

MILITARY

MIL-L-19140	- Lumber and Plywood, Fire Retardant Treated.
MIL-P-24441	- Paint, Epoxy-Polyamide, General Specification for.
MIL-P-24441/1	- Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type I.
MIL-E-24635	- Enamel, Silicone Alkyd Copolymer

STANDARDS AND COMMERCIAL ITEM DESCRIPTIONS

FEDERAL

FTMS-141	- Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing.
FED-STD-313	- Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.
FED-STD-595	- Colors Used in Government Procurement.
CID-A-A-1045	- Grain Abrasive

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MIL-STD-129	- Marking for Shipment and Storage.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY (USEPA)

40 Code of Federal Regulations (CFR) CH. 1, Part 60, Appendix A, Method 24 - Determination of Volatile Matter Content, Water Content, Density, Volume Solids and Weight Solids of Surface Coatings.

40 CFR part 261, Appendix II Toxicity Characteristic Leaching Procedure (TCLP)

Test method for evaluating solid waste physical/chemical methods, SW-846 (NSN-955-001-00000-1)

Federal Register, Volume 47, Number 103, Appendix A, pages 23376-23389, May 7, 1982

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Code of Federal Regulations (CFR)

29 CFR 1910.1200, 1915.99, 1917.28, 1918.90, and 1926.59 - Hazard Communication.

49 CFR 171-178 -Regulations for the Transportation of Hazardous Materials.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

STATE OF CALIFORNIA DEPARTMENT OF HEALTH SERVICES

California Administrative Code Title 22 Division 4, Chapter 30, Minimum Standards for Management of Hazardous and Extremely Hazardous Waste, Section 66699, Persistent and Bioaccumulative Toxic Substance and Section 66700 Waste Extraction test, Pg 679-681 (Register 90, Nos 38-41; 10-12-90).

(Application for copies should be addressed to the Hazardous Material Laboratory, state of California, Dept of Health Services, 2151 Berkeley Way, Berkeley, CA 94704)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI).

ANSI Z 129.1 Precautionary Labelling of Hazardous Industrial Chemicals.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 523 - Standard Test Method for Specular Gloss. (DoD adopted)
D 562 - Standard Test Method for Consistency of Paints Using the Stormer Viscometer. (DoD adopted)
D 609 - Standard Practice for Preparation of Cold-Rolled Steel Panels for Testing Paint, Varnish, conversion Coatings and Related Coatings Products
D 660 - Standard Test Method for Evaluating Degree of Checking of Exterior Paints. (DoD adopted)
D 661 - Standard Test Method for Evaluating Degree of Cracking of Exterior Paints. (DoD adopted)
D 714 - Standard Test Method for Evaluating Degree of Blistering of Paints. (DoD adopted)
D 823 - Standards Practice for Producing Films of Uniform Thickness of Paint, Varnish and Related Products on Test Panels. (DoD adopted)
D 1014 - Standard Test Method for Conducting Exterior Exposure Tests of Paints on Steel. (DoD adopted)
D 1210 - Standard Test Method for Fineness of Dispersion of Pigment-Vehicle Systems. (DoD adopted)
D 1654 - Standard Method for Evaluation of Painted or Coated Specimens Subjected to Corrosive Environments. (DoD adopted)
D 2243 - Standard Test Method for Freeze-Thaw Resistance of Water-Borne Coatings. (DoD adopted)
D 2244 - Standard Method for Calculation of Color Differences from Instrumentally Measured Color Coordinates. (DoD adopted)
D 2697 - Standard Test Method for Volume Nonvolatile Matter in Clear Pigmented Coatings.
D-2794 - Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact).
D 2805 - Standard Test Method for Hiding Power of Paints by Reflectometry.
D 3168 - Standard Practice for Qualitative Identification of Polymers in Emulsion Paints.

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D 3273 - Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber. (DoD adopted)
D 3274 - Standard Test Method of Evaluating Degree of Surface Disfigurement of Paint Films by Microbial (Fungal or Algal) Growth or Soil and Dirt Accumulation. (DoD adopted)
D 3278 - Standard Test Methods for Flash Point of Liquids by Setaflash Closed-Cup Apparatus. (DoD adopted)
D 3335 - Standard Test Method for Low Concentrations of Lead, Cadmium, and Cobalt in Paint by Atomic Absorption Spectroscopy. (DoD adopted)
D 3359 - Standard Methods for Measuring Adhesion by Tape Test. (DoD adopted)
D 4214 - Standard Test Method for Chalking of Exterior Paint Films.
D 4400 - Standard Test Method for Sag Resistance of Paints Using a Multinotch Applicator.
E 97 - Standard Test Method for Directional Reflectance Factor, 45-Deg 0-Deg, of Opaque Specimens by Broad-Band Filter Reflectometry. (DoD adopted)
F 718 - Standard for Shipbuilders and Marine Paints and Coatings Product/Procedure Data Sheet.(DoD adopted)
G 53 - Standard Practice for Operating Light- and Water-Exposure Apparatus (Fluorescent UV-Condensation Type) for Exposure of Nonmetallic Materials. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The enamels furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) at the time of award of contract (see 4.3, 6.2 and 6.4).

3.2 Material safety data sheet (MSDS). The contracting activity shall be provided a material safety data sheet at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.6).

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3.3 Directions for use. The contractor shall provide written directions on each container for the surface preparation, mixing, and applying of the enamel supplied and this direction shall include all information necessary to comply with OSHA Hazard Communication Standard and FED-STD-313. In addition, the contractor shall provide an ASTM F 718 data sheet which shall separately detail requirements for small unit (metric and English equivalents of pint, quart, gallon) and large unit (5-gallon) containers.

3.4 Hazardous material. Manufacturers shall prepare container hazardous chemical warning label for the paint in accordance with the requirements of 29 CFR Parts 1910.1200, 1915.99, 1917.28, 1918.90, and 1926.59 - OSHA Hazard Communication Standard, Final Rule (see 5.1). In choosing the specific ingredients to manufacture the paint, the manufacturer shall certify in writing to Naval Sea Systems Command (SEA 03Q2) that the paint offered for delivery does not contain the following materials greater than of 0.05% by weight of either the total formulated paint or in the dry film formed by the paint: Benzene, Toluene, chlorinated solvents, esters or ethers derived from ethylene glycol, hydrolyzable chlorine derivatives, coal tar and coal tar derivatives.(see 6.3)

3.4.1 Toxicity of product/formulation. The materials shall have no adverse effect on the health of personnel when used for its intended purpose. When evaluated in accordance with 4.7, the ingredients used in the formulation of this product, unless specific maximum levels are cited elsewhere in this specification, shall have no known or suspect carcinogenic materials, as identified in paragraph 3.4.
This specification places severe restrictions on the presence of the following substances in this product.

a. Substances which are identified as specific "target compounds". Target compounds in this specification are as follows: Benzene, Toluene, chlorinated solvents, esters or ethers derived from ethylene glycol, hydrolyzable chlorine derivatives, coal tar and coal tar derivatives.

b. Substances which are regulated as carcinogens by the Occupational safety and Health Administration (OSHA).

c. Substances which are considered to be known, probable, or possible human carcinogens, by the International Agency for Research on Cancer (IARC) (i.e.., Group 1, 2A, and 2B).

d. Substances which are known or reasonable anticipated to be carcinogenic, as found in the National Toxicology Program's latest annual report.

e. Substances identified as a Navy Occupational Reproductive Chemical Stressor, Appendix 28-A, (See 6.7)

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f. For Substances meeting one or more of the previous criteria (a) thru (e) , the following limitations apply:

(1) These substances shall not be intentionally added to any formulation intended to comply with this specification.

(2) Where the substances may find its way into a formulation, as a result of being present as a "trace" or "impurity" in another ingredient, concentrations of the substances shall be maintained [as low as possible].

(3) Under no circumstances shall the substance in the product offered for delivery (either in the total formulated product or in the dry film formed by the product) exceed limits (as delineated in this specification).

g. Substances that have been identified by Federal EPA or State regulatory agencies as hazardous air pollutants (HAPS).

Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as advisor to the contracting activity.

3.4.2. Crystalline silica content. When evaluated in accordance with 4.7 the coating of shall not contain crystalline silica in excess of 0.1 % by weight of the dry paint.

3.4.3. Hazardous air pollutants (HAPs). The content of HAPs solvents in the total enamel shall not be greater than the weight percent (%WT) values listed in table I when tested in accordance with 4.6.14.3.

Table I. Hazardous solvent content1/.

Hazardous Solvent in total enamel	Maximum, %WT
Benzene	0.05
Chlorinated solvent(s), total	0.05
Carbon tetrachloride	
Chloroform (Trichloromethane)	
Methylene chloride (Dichloromethane)	
Tetrachloroethylene (Perchloroethylene)	
1,1,1-Trichloroethylene (Methyl chloroform)	
Trichlorothylene	
Ethyl benzene	0.05
Methyl, ethyl and butyl mono-ethers of Ethylene glycol or the acetates, total (Methyl, Ethyl and Butyl Cellosolves and acetates)	0.05
Methyl ethyl ketone (MEK)	0.05
Methyl isobutyl ketone (MIBK)	0.05
Toluene	0.05
Xylene (all forms), total	0.1

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3.5 Composition. The contractor is given his choice in the formulation of the enamels provided the end product conforms to all requirements of this document (see 4.7). However, Contractor shall assure composition conforms to the requirements of 3.4 and that soluble metal content and total metal content of the enamels shall be not greater than the values listed in Tables II and III when tested in accordance with 4.6.14.

TABLE II. Soluble metals content1/.

Requirement	Maximum
Metals content (soluble), mg/L	
Antimony and/or its compounds	15
Arsenic and/or its compounds	5
Barium and/or its compounds (excluding barite)	100
Beryllium and/or its compounds	0.75
Cadmium and/or its compounds	1
Chromium (VI) compounds	5
Chromium and/or chromium (III) compounds	560
Cobalt and/or its compounds	10
Copper and/or its compounds	0.5
Fluoride salts	180
Lead and/or its compounds	0.5
Mercury and/or its compounds	0.2
Molybdenum and/or its compounds	350
Nickel and/or its compounds	10
Selenium and/or its compounds	1
Silver and/or its compounds	5
Thallium and/or its compounds	7
Tungston and/or its compounds	10
Tuntalum and/or its compounds	10
Vanadium and/or its compounds	24
Zinc and/or its compounds	250

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TABLE III. Total metals content1/.

Requirement	Maximum
Metals content (total), weight percent	
Antimony and/or its compounds	0.05
Arsenic and/or its compounds	0.05
Barium and/or its compounds (excluding barite)	1.00
Beryllium and/or its compounds	0.0075
Cadmium and/or its compounds	0.01
Chromium (VI) compounds	0.005
Chromium and/or chromium (III) compounds	0.05
Cobalt and/or its compounds	0.10
Copper and/or its compounds	0.25
Fluoride salts	1.80
Lead and/or its compounds	0.005
Mercury and/or its compounds	0.002
Molybdenum and/or its compounds	0.35
Nickel and/or its compounds	0.20
Selenium and/or its compounds	0.01
Silver and/or its compounds	0.05
Thallium and/or its compounds	0.07
Tungston and/or its compounds	0.10
Tuntalum and/or its compounds	0.10
Vanadium and/or its compounds	0.24
Zinc and/or its compounds	0.50

1/ By data certification

3.6 Material requirements. The enamel shall consist of pigments, additives, and acrylic emulsion resins, so combined as to produce a ready-for-use material meeting all the requirements of this document. Infrared analysis as specified in table V shall show the resin to be 95 percent unmodified acrylic.

3.7 Quantitative requirements. The enamel shall conform to the quantitative requirements specified on table IV and as follows when tested as specified in section 4.

3.7.1 Color. The color shall be characteristic of the pigments used and shall match the following as specified (see table VI and 4.6.5):

(a). Colors defined by Tristimulus values shall match the following when measured in accordance with 4.6.7:

	Tristimulus values		
Colors defined by tristimulus values	L	a	b
Navy Haze Gray No. 27	+56.0	-1.83	-1.37

(b). All other colors shall match the FED-STD-595 number specified in Table VI when measured in accordance with 4.6.5.

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TABLE IV. Quantitative requirements of all types and classes, unless specifically indicated.

	Minimum	Maximum
Contsistancy, Krebs-Stormer, shearing rate, equivalent krebs units	------	100
Fineness of grind, Hegman	6	--
Flashpoint, degrees Celsius(°C) (Fahrenheit (°F)	38(100)	--
VOC, grams per waterless liter (LB/gal)		
Type I	--	340 (2.8)
Type II	--	250 (2.1)
Drying time, hours		
Set-to-touch	--	2
Dry hard	--	8
Dry-to-recoat	--	16
Gloss, 60 degrees specular		
Class 1	85	--
Class 2	45	60
Class 3	15	30
Class 4	5	15
Volume solids (percent) ASTM D 2697	35	---
Coarse particles and skins, percent by weight	--	0.2
Flexibility, 1 mil dry film, 0.25-inch mandrel	pass	--
Impact resistance, 1 mil dry film, 3.6 inch-pounds		
Direct	no damage 1/	--
Reverse	no damage 1/	--
Lead, percent by weight of total solids	--	0.005 2/
Dry film contrast ratio, percent	0.98	--
Sag resistance, anti-sag index	7	--
Contrast ratio		
Gray and black colors	0.99	--
White colors	0.94	--
Red, yellow and orange	0.93	--
All other Colors	0.96	--
Soluble and total metal content	--	3/

1/ See 3.8.11
2/ The allowable levels of lead and lead compounds (total).005 shall be maintained at this level or lower
3/ See 3.5

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3.7.1.1 Color deviation. The color deviation of the paint shall be obtained as specified in 4.6.5. The measured color deviation terms [Delta (Δ) E, Δ A, Δ B, Δ L] from the appropriate Haze Gray No. 27 [see 3.7.1 (a)] or specified FED-STD-595 color card in Commission Internationale de l'Eclairage (CIE; International Commission on Illumination) LAB units shall be not greater than the CIELAB units listed below as either positive or negative (+ or -) values:

	Color Deviation values			
Colors defined by tristimulus values	Δ E	Δ A	Δ B	Δ L
Navy Haze Gray No. 27	0.5	0.3	0.3	0.3

	Color Deviation values			
FED-STD-595 COLOR No.	Δ E	Δ A	Δ B	Δ L
26173, 26373	0.5	0.3	0.3	0.3
36173, 36270, 36373	0.5	0.3	0.3	0.3
All other colors	1.8	1.0	1.0	1.0

3.7.2 Accelerated weathering. When prepared as specified in 4.6.11.1 and tested in accordance with 4.6.13, films of the enamels shall meet the following requirements:

(a) be not rated less than 9 (1 percent) for checking (ASTM D 660)

(b) be not rated less than 9 for cracking (ASTM D 661)

(c) be not rated less than 10 for blistering (ASTM D 714)

(d) gloss retention shall be not less than 65 percent of that measured prior to weathering (ASTM D 523)

(e) the CIELAB color difference (delta E), delta L, delta a, and delta b units, shall be not greater than 1.0 CIELAB units from the original color values measured before testing (ASTM D 2244)

(f) the degree of chalking shall not be more than 8 when tested in accordance with D 4214.

3.7.3 Volatile organic content. The VOC of the enamel shall be not greater than the quantities allowed in table IV when determined as specified in 4.6.8 (see 6.3).

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3.7.4 Long term exterior exposure. When tested in accordance with 4.6.15, films of the enamel shall meet the following requirements:

(a) 60° Gloss must be less than 50% of that measured prior to testing,

(b) Chalking shall be rated not less than number 8 of ASTM D 4214.

(c) Checking shall be rated not less than 9 (1 percent [%]).

(d) Cracking shall be rated not less than 9.

(e) Blistering shall be rated not less than 10.

(f) When CIE color values of the exposed test panels are compared to the original CIE color values of the test panel, the ASTM D 2244 color difference calculated values shall conform to the following values:

(1) Delta E shall be not greater than 1.0 CIELAB units.

(2) Delta a shall be not greater than positive or negative (+/-) 1.0 CIELAB unit.

(3) Delta b shall be not greater than +/- 1.0 CIELAB unit.

(4) Delta L shall not exceed a +/- 1.0 CIELAB unit.

3.8 Qualitative requirements (all types). The enamel shall meet the following qualitative requirements specified herein.

3.8.1 Condition in the container. When tested in accordance with method 3011.2 of FTMS-141, the enamel shall be ready mixed and shall show no hard pigment settling, phase separation, evidence of biological growth, livering, skinning, putrefaction, corrosion of the container, pigment float, grit or seeding particles, or persistent foam. Any settled or floating pigment shall be easily and completely reincorporated by mechanical shaking within 5 minutes to a smooth and uniform condition, free from persistent foam. When tested in accordance with 4.6.5. the CIELAB color difference of the mixed enamel shall meet the requirements of 3.7.1.1. After a standing period of 2 hours without further mixing, the CIELAB color difference of the enamel shall again be tested in accordance with 4.6.5. The CIELAB color difference measured for 2 hours old mixed enamel shall be not greater than 0.5 CIELAB units from the color values measured after stirring, but before the 2 hour standing period. After standing for 2 hours there shall be no settling or pigment float.

3.8.2 Resistance to biological growth. When tested as specified in 4.6.9, the enamel film shall have a surface disfigurement rating of greater than or equal to 8 when evaluated against adjunct no. 12-432740-00 as specified in ASTM D 3274.

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3.8.3 Freeze-thaw stability. When tested as specified in 4.6.10, the enamel shall be free of livering, curdling, hard-dry caking, pigment settlement or pigment float and tough gummy sediments. The enamel shall mix readily with a paddle or mechanical mixer within 5 minutes to a smooth, uniform state, free of grits, seed, tough or gummy sediment, skins, pigment settling and persistent foam. The tested enamel shall show no flocculation and its consistency shall change not greater than plus or minus 8 KU compared to the paint in an original, unopened container in the as-received condition. Additionally, the tested enamel shall meet the requirements of gloss retention and color change required by 3.7.2.

3.8.4 Storage stability.

3.8.4.1 Shelf life. When tested as specified in 4.6.1.1 one year after date of manufacture an original, unopened full container of enamel shall be free of grits, seeds tough or gummy sediment, skins, pigment settling, pigment float, and persistent foam. The tested enamel shall mix readily to a smooth, uniform state with a paddle or mechanical mixer within 5 minutes and show no flocculation and its consistency shall change no greater than plus or minus 8 KU as compared to the paint when first delivered. Additionally, the tested paint shall meet the requirements of 3.7.2. When tested in accordance with 4.6.5, the CIELAB color difference of the mixed enamel shall be not greater than 1.0 CIELAB units from the original color values measured prior to the shelf life test. The mixed enamel shall also meet the requirements of 3.7.2. The one year paint shelf life is non-extendable.

3.8.4.2 Partially full container. After exposure as specified in 4.6.1.2, the enamel shall show no skinning, pigment float, livering, curdling, hard dry caking, or tough or gummy sediment. The enamel shall mix readily with a paddle or mechanical mixer within 5 minutes to a smooth, uniform state, free of grits, seeds, lumps, and skins. When tested in accordance with 4.6.5, the CIELAB color difference of the mixed enamel shall be not greater than 1.0 CIELAB units from the original color values measured prior to the shelf life test. The mixed enamel shall also meet the requirements of 3.7.2.

3.8.4.3 Accelerated storage stability. After storage at 60 degrees Celsius (°C) (140°F) for 30 days as specified in 4.6.1.3, the unopened, original container of enamel shall be redispersable by mechanical shaking within 5 minutes to a uniform condition and shall meet the requirements specified in 3.8.1, 3.8.5, 3.8.6, and 3.8.7. When tested in accordance with 4.6.5, the CIELAB color difference of the mixed enamel shall be not greater than 1.0 CIELAB units from the original color values measured prior to the shelf life test. The mixed enamel shall also meet the requirements of 3.7.2. when the color difference is calculated using as the original color that obtained for the one year storage test sample.

3.8.5 Spraying properties. When tested as specified in 4.6.2, the enamel shall spray satisfactorily and shall show no running, sagging, or streaking. The dried film shall show no blooming, blushing, bubbling, cratering, dusting, floating, fogging, hazing, mottling, orange peel appearance, pinholing, seeding, silking, or streaking in accordance with method 4331.1 of FTMS-141.

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3.8.6 Brushing properties. When tested as specified in 4.6.3, the enamel shall brush easily and have good flowing and spreading qualities and shall dry to a uniform film. The dried film shall show no running, sagging, or streaking. The dried film shall show no runs, blooming, blushing, bubbling, cratering, dusting, floating, fogging, hazing, mottling, orange peel appearance, pinholing, seeding, silking, or streaking in accordance with method 4321.2 of FTMS-141.

3.8.7 Rolling properties. When tested as specified in 4.6.4, the enamel shall roll easily and have good flowing and spreading qualities. The dried film shall show no running, sagging, blooming, blushing, bubbling, cratering, dusting, floating, fogging, hazing, mottling, orange peel appearance, pinholing, seeding, silking, or streaking in accordance with method 4335 of FTMS-141.

3.8.8 Knife test. When tested as specified in 4.6.7, a film of enamel shall adhere tightly to the primer. It shall be difficult to furrow off with the knife and shall not flake, chip or powder. The enamel shall not show brittle or rubbery characteristics; the knife cut shall show beveled edges.

3.8.9 Primer compatibility.

3.8.9.1 Dry adhesion. When tested as specified in 4.6.11.1, films consisting of primer and enamel shall show no blistering, wrinkling or any other evidence of separation. The adhesion by tape test shall be rated not less than 4A. The enamel shall adhere tightly to the primer. Using a scalpel, the enamel shall not lift from the primer without leaving some enamel attached to the primer.

3.8.9.2 Wet adhesion. When tested as specified in 4.6.11.2, films consisting of primer and enamel shall show no blistering, wrinkling or any other evidence of separation. The adhesion by tape test shall be rated not less than 4A. The enamel shall adhere tightly to the primer. Using a scalpel, the enamel shall not lift from the primer without leaving some enamel attached to the primer

3.8.10 Enamel recoatability. When tested as specified in 4.6.12, films of enamel shall show no blistering, wrinkling or other evidence of intercoat separation. Films shall be difficult to separate from one another. Using a scalpel, there shall be no lifting between coats without leaving coating on the overcoated film.

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3.8.11 Impact resistance The enamel when tested in accordance with 4.6.16 shall not exceed the maximum percent area increase accommodated by the enamel without cracking or other failure as specified for each spherical segment in accordance with the following:

Spherical Segment	End	Base Diameter	Segment Radius	Segment Elevation	Max % Area Increase
1	A	0.375	0.194	0.146	60
2	A	0.375	0.208	0.119	40
3	A	0.375	0.252	0.084	20
4	A	0.375	0.326	0.059	10
5	B	0.375	0.440	0.042	5
6	B	0.375	0.676	0.027	2
7	B	0.375	0.947	0.019	1
8	B	0.375	1.332	0.013	0.5

3.8.12 Directions for use. The manufacturer must provide written directions for the mixing and applying of the enamel supplied and this direction shall include all information necessary to comply with OSHA Hazard Communication Standard and FED-STD-313. In addition, the manufacturer must prepare an ASTM F 718 data sheet which separately details requirements for small unit (pint, quart, gallon) and large unit (5 gallon) containers.

3.8.13 Compatibility test. When tested in accordance with 4.6.17, the dried enamel film shall not show any visual difference in color and gloss, and no streaks, craters, or pigment floating.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection (see 6.4) shall be conducted at a commercial or manufacturer's laboratory approved by Naval Sea Systems Command (NAVSEA). Test shall be monitored by the Defense Contract Management Area Operations (DCMAO). Qualification tests shall consist of all tests specified in table V. If no test paragraph is listed in table V, the test shall be conducted in accordance with the method listed and results shall comply with the requirement paragraph listed. A test report detailing the results of any required testing performed by the manufacturer, his representatives or at his behest by employed agents shall be provided by the manufacturer to the contracting officer, his agent or the qualifying activity, as appropriate (see 6.3).

4.3.1 Qualification samples. For qualification testing specified in 4.3, a sample of enamel conforming to Navy Haze Grey no. 27, FED-STD-595 color number 26270 shall be submitted.

4.3.1 Extension of qualification. Approval of qualification for Navy haze gray to match color 26270 of FED-STD-595 shall constitute approval for other colors of the same type and class. Approval of qualification for a type II class enamel shall also constitute approval of qualification of that enamel for type I of the same class. Enamel submitted for qualification for a type II, but passes the VOC requirement for type I shall be qualified for the same type I class provided that the enamel passes all other requirements for that type and class. Enamel submitted for qualification for a type II class, but that does not pass the VOC requirement for type II, but passes the VOC requirement for type I, shall be qualified for the same type I class provided that the enamel passes all other requirements for that type and class.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the table IV tests marked with a 1/ (see table IV footnote). Tests shall be performed on each lot offered for delivery (see 6.3).

4.4.1 Lot. For the purpose of quality conformance inspection and test sampling, a lot is defined as all the enamel (in U.S. gallons) of the same type, class and color, from a single uniform batch, produced and offered for delivery at one time. The addition of any substance to a batch shall constitute a new lot.

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4.4.1.1 Batch. A batch shall consist of a quantity of an individual finished paint manufactured at the same time from the same units (sacks, cans, barrels, etc.) of ingredients.

4.4.2 Sampling for quality conformance inspection. As a minimum, the contractor shall select at random two samples of sufficient size for testing. One sample shall be taken from the first 10% of production from the lot, The other sample shall be taken from the last 10% of production from the lot. The samples shall be inspected and tested in accordance with table V.

4.4.3 Lot rejection. Failure of any sample to pass any inspection or test specified in table V of this specification shall be cause for rejection of the entire lot represented by the sample. Rejection of a lot shall require that corrective action be implemented by the contractor.

4.5 Test conditions. Unless otherwise specified, the routine and referee testing conditions shall be in accordance with FTMS-141. The term referee condition shall mean a temperature of 23 ± 1°C (73 ± 2°F) and a relative humidity of 50 ± 4 percent. A dry film thickness of 3.8 to 5.1 mm (0.0015 to 0.002 inches ((1.5-2.0 mils)) shall be used whenever film thickness is required in any test unless otherwise specified by the test.

4.5.1 Test panels and surface preparation.

4.5.1.1. Test panels. Test panels shall consist of two types. The two types are:

(1) Aluminum: Nominal 150 x 300 x 6 millimeter (mm) (6 x 12 x 0.25 inch [in]) plate aluminum IAW QQ-A-250/4 (0 temper). Aluminum plates shall be abrasive blasted with aluminum oxide that meets Commercial Item Description A-A-1045, Type I, Class 1 and solvent cleaned to provide a uniform, clean rough surface with a profile of 0.050 to 0.076 mm (0.002 to 0.003 in) (nominal) on both sides.

(2) Glass: Nominal 150 x 300 x 6 millimeter (mm) (6 x 12 x 0.25 inch [in]) glass, ground or abrasive blasted with aluminum oxide that meets Commercial Item Description A-A-1045, Type I, Class 1 and solvent cleaned to provide a uniform, clean, rough surface on one side.

Primers specified in 4.5.1.2 shall be applied to the prepared side(s) of the test panel. Cure for 24 hours at room temperature and then 1 hour at 40 to 50°C (104 to 122°F) unless otherwise specified herein.

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4.5.1.2 Test primers. The following primers shall be used:

(a) MIL-P-24441/1.
(b) A Naval Sea Systems Command (NAVSEA 03M1) approved inorganic zinc-rich primer.
(c) TT-P-645

4.6 Test procedures (all types). Test procedures shall be as listed in table IV, and as otherwise specified herein.

4.6.1 Storage stability.

4.6.1.1 Shelf life. When allowed to remain undisturbed for 2 years under ambient conditions (see 4.5), the enamel shall conform to the requirements in 3.8.4.1.

4.6.1.2 Partially full container. Skinning shall be determined after 48 hours in accordance with method 3021.1 of FTMS-141, except that a 3/4-filled, 1/2-pint, resin-lined, multiple friction top can shall be used. The can shall be resealed and aged for 30 days at 60°C (140°F) and examined for compliance with 3.8.4.2.

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TABLE V. Test procedures.

Characteristic	Requirement	Applicable test method FTMS-141	ASTM test method	Test
Material requirements	3.6	-----	D 3168	-----
Condition in container 1/	3.8.1	3011.2	-----	-----
Storage stability	3.8.4	3021.1	-----	4.6.1
Spraying properties 1/	3.8.5	4331.1	-----	4.6.2
Brushing properties 1/	3.8.6	4321.2	-----	4.6.3
Rolling properties 1/	3.8.7	4335	-----	4.6.4
Color/color deviation 1/	3.7.1	-----	E 97, D 2244	4.6.5
Flexibility	Table IV	6221	-----	4.6.6
Knife test	3.8.8	6304.1	-----	4.6.7
Consistency 1/	Table IV	-----	D 562	4.6.10
Sag resistance 1/	Table IV	-----	D 4400 Method A	-----
Fineness of grind 1/	Table IV	-----	D 1210	-----
Flash point 1/	Table IV	-----	D 3278	-----
VOC 1/ 4/	Table IV, 3.7.3	-----	-----	4.6.8
Drying time 1/	Table IV	4061.2	-----	-----
Gloss, 60 degrees specular 1/2/	Table IV	-----	D 523	4.6.18
Coarse particles 1/	Table IV	4092.1	-----	-----
Impact resistance	Table IV	------	D 2794	4.6.16
Lead content	Table IV	-----	D 3335	-----
Resistance to biological growth	3.8.2	-----	D 3274, D 3273	4.6.9
Freeze-thaw stability	3.8.3	-----	D 2243	4.6.10
Primer compatibility	3.8.9	6301.2	D 3359 Method A	4.6.11
Enamel recoatability	3.8.10	-----	-----	4.6.12
Accelerated weathering	3.7.2	-----	-----	4.6.13
Contrast ratio 3/	Table IV	-----	D 2805	4.6.19
Soluble and total metal content 4/	3.5	-----	-----	4.6.14
Long term exterior exposure	3.7.4.	----	-----	4.6.15
Compatibility	3.8.13	-----	-----	4.6.17

1/ Required for quality conformance inspection.
2/ Make a 3 mil wet film thickness drawdown on glass, dry for 168 hours at ambient laboratory conditions before performing test.
3/ Spreading rate shall be such as to yield a dry film thickness of 3 to 4 mils.
4/ By data certification

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4.6.1.3 Accelerated storage stability. After exposure to a temperature of 60°C (140°F) continually for 30 days, a new unopened full 1-quart can of enamel shall be examined for compliance with 3.8.4.3.

4.6.2 Spraying properties. The enamel shall be sprayed in an "as-received" condition on a ground glass panel (see 4.5.1.1) to a wet film thickness of 0.002 to 0.004 inch (46 to 92 micrometer (µm)). The spray gun distance shall be 6 to 12 inches for conventional spraying and 12 to 15 inches for airless. The panel shall be observed for spraying properties in accordance with method 4331.1 of FTMS-141 for compliance with 3.8.5.

4.6.3 Brushing properties. The brushing properties of the enamel shall be determined in accordance with method 4321.2 of FTMS-141 for compliance with 3.8.6.

4.6.4 Rolling properties. The rolling properties of the enamel in an "as-received" condition shall be determined in accordance with method 4335 of FTMS-141 for compliance with 3.8.7.

4.6.5 Color testing.

4.6.5.1 Instrumental color deviation determination. Test specimens shall be prepared in accordance with ASTM D 823 Test Methods C or E using a nominal 0.006 in blade film applicator on to a clear plate glass of not less than 1/4 in (nominal) thickness which has been ground to a uniform finish with 1F carborundum. The color deviation shall be determined in accordance with ASTM D 2244 using a instrument having a D_{65} light source (10 degree observer) or C (2 degree observer), a 45 degree illumination angle, a 0 degree viewing angle. The instrument shall be calibrated in accordance with manufacturer instructions and shall be demonstrated to read the color of National Institute for Science and Technology (NIST [formerly National Bureau of Standards; NBS]) traceable standards with color deviations not greater than allowed in 3.3. After calibration of the instrument, measure the CIE color values of not less than two FED-STD-595 color cards of the color being procured which were received from the Government not greater than 1 year prior to the date of this use. Making not less than three readings per card taken at one inch intervals on the long axis, one inch from the edge. Calculate E for each color card. Determine the mean values of E and the CIE color values for the color cards. If any individual CIE color value differs from the mean color value by more than 0.05, use a different color card and recalculate the mean values. These mean values will be used to calculate the color difference values. Measure the CIE color values of the test enamel using the values of three readings taken at one inch intervals on the long axis, one inch from the edge. Calculate the mean CIE color values of the test enamel. Calculate the color difference terms. If using an instrument that automatically calculates the color differences, enter the color card mean values as the base line for the color difference calculations. The results of the color difference calculations shall be in accordance with the requirements of 3.7.1 or table VI, as applicable.

4.6.7.2 Referee color matching. If the enamel measured in accordance with 4.6.7.1 does not match the color card when visually examined, the discrepancy shall be resolved by an independent laboratory certified to the NIST National Volunteer Laboratory Accreditation Program for paint testing. When tested in accordance with ASTM D 823, the test enamel, prepared as in 4.6.7.1, shall visually match the FED-STD-595 color card that most closely corresponds to the mean values calculated in 4.6.7.1.

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4.6.6 Flexibility. The flexibility of the enamel shall be determined in accordance with method 6221 of FTMS-141. A 2-inch wide film of enamel shall be applied using a film applicator that will give a dry film thickness (DFT) of 0.0009 to 0.0011 inch (23 to 28 µm) on a smooth finished tinplate steel panel prepared in accordance with method 2011.2 of FTMS-141 using a mixture of 50:50 aliphatic naphtha and propylene glycol monoethyl ether. The panel shall be prepared from new, cold-rolled, rust-free, carbon steel 0.010 ± 0.001 inch (25.4 ± 2.5 µm) thick with a Rockwell 15-T maximum hardness of 82 and finished with a surface roughness of 8 to 12 µm. After application, the panel shall be air-dried in a horizontal position for 24 hours at 23 ± 1.1°C (73.4 ± 2°F). The panel shall be conditioned for 1/2 hour under standard test conditions. The panel shall be bent over a 1/4-inch (6.35-millimeter (mm)) mandrel. The panel shall be examined for cracks over the area of the bend in a strong light for compliance with table IV.

4.6.7 Knife test. The knife test shall be conducted in accordance with method 6304.1 of FTMS-141, using the flat portion of the panel from the flexibility test (see 4.6.6). The test results shall conform to 3.8.8.

4.6.8 Volatile organic content. The VOC in the enamel shall conform to the requirements of table III, when determined in accordance with EPA 40 CFR CH.1, part 60, appendix A, method 24.

4.6.9 Resistance to biological growth. The biological growth on the surface of the enamel shall be determined in accordance with ASTM D 3273. The extent of surface disfigurement shall be evaluated in accordance with ASTM D 3274. Fungal mycelium and spores, slime, and dirt or soil accumulation shall be considered disfiguring agents in the evaluation. The rating obtained shall be as specified in 3.8.2.

4.6.10 Freeze-thaw stability. Enamel shall be tested in accordance with ASTM D 2243, except after completion of the exposure and room temperature equilibration, the exposure shall be repeated twice before proceeding to the examination (three exposure cycles to 0°F and three equilibrations). Consistancy shall be determined in accordance with ASTM D 562; contrast ratio, gloss, and reflectance shall be determined as specified in table IV; and characteristics shall comply with the requirements of table III (see 3.7.2 and 3.8.3).

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4.6.11 Primer compatibility.

4.6.11.1 Dry adhesion. The 14 test panels shall be as specified in 4.5.1.1. Six panels shall be primed with two coats of an epoxy-polyamide material in accordance with MIL-P-24441 and MIL-P-24441/1. Four panels shall be primed in accordance with TT-P-645. Four panels shall be prepared with a NAVSEA 03M1 approved inorganic zinc-rich primer in accordance with manufacturer's instructions. Two panels, each prepared with MIL-P-24441/1, shall dry for 6, 24, 168 hours, respectively, at standard test conditions (see 4.5) before the application of two spray coats of enamel conforming to this specification. Film thickness shall be 2 to 4 mils for MIL-P-24441/1 and 1.5 to 2.0 mils for TT-P-645, respectively. The panels dried for 168 hours, prior to application of the enamel, shall receive a tack coat of the MIL-P-24441/1 primer (a wet film application of 1 to 2 mils). Two panels, each to which primer in accordance with TT-P-645 is applied, shall be dried for 24 and 168 hours, respectively, at standard test conditions (see 4.5) and overcoated with enamel conforming to this specification in like manner as the panels treated with the epoxy-polyamide material. Panels of the NAVSEA 03M1 approved inorganic, zinc-rich primer shall be overcoated in the same manner as the TT-P-645 test panels. Overcoated primed panels shall be dried for 168 hours under standard test conditions (see 4.5). Panels shall be tested as specified in 4.6.13. The tape test shall be performed in accordance with ASTM D 3359, method A, except that the tape used shall be commercially available masking tape. Following the tape test, lifting characteristics shall be determined between the enamel and the primer at the scribe and at a distance up to 1 inch perpendicular to the scribe using the aid of a microscope. The test results shall conform to the requirements of 3.8.9.1.

4.6.11.2 Wet adhesion. Application of primers and enamel shall be made as specified in 4.6.11.1 and the systems shall be dried as previously indicated. At the conclusion of the drying cycle, the panels shall be scribed using a tool in accordance with ASTM D 1654 and immersed in distilled water at standard test conditions (see 4.5) for 24 hours. While immersed, the panels shall be examined for evidence of blistering, wrinkling, or other evidence of adhesion failure of the test enamel to the primers. Test for wet adhesion shall be in accordance with method 6301.2 of FTMS-141, except the tape used shall be commercially available masking tape. Following the tape test, lifting characteristics shall be determined between the enamel and the primer at the scribe and at a distance up to 1 inch perpendicular to the scribe using the aid of a microscope. The test results shall conform to the requirements of 3.8.9.2.

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4.6.12 Enamel recoatability. Ten test panels of aluminum having nominal dimensions of 150x300x3 millimeters (mm) (6x12x1/8 inches[in]) shall be prepared. Test requires 10 test panels. Test panels shall be degreased in accordance with methods B, C or D of ASTM D 609. The degreased test panels shall be abrasive blasted with new, clean aluminum oxide to provide a profile pattern of 0.0253 to 0.0506 mm (0.001 to 0.002 in) (average). All test panels shall be primed with 2 spray applied coats of MIL-P-24441 Type III or IV F.150 Green primer. Each coat of MIL-P-24441 shall be applied to 0.051 to 0.101 mm (0.002 to 0.004 in) dry film thickness and allowed to cure 24 hours at ambient laboratory conditions before being overcoated. Two coats of the test enamel shall be spray applied on the MIL-P-24441 F.150 coated panels to obtain a final dry film thickness of the test topcoat of 101 to 151 mm (0.004 to 0.006 in). Allow not less than 24 hours dry time between application of the coats of test topcoats at ambient laboratory conditions. The completed painted test panels shall be cured for not less than 7 days at ambient laboratory conditions before proceeding to the mounting on the test rack. Spraying properties shall be in accordance with 3.8.5, as appropriate. Test specimens shall be mounted on a test fence having a 45° south facing rack at a distance of not greater than 30 meters (100 feet) of an ocean high tide line. The exposure shall be for a period of not less than one year (365 days) in accordance with paragraph 6 of ASTM D 1014. The fence shall be at the Naval Research Laboratory, Marine corrosion Laboratory, Key West, Florida, USA. After the exposure period of one year (365 days), the panels shall be returned to the laboratory. The panels shall be rinsed with cold tap water at nominal tap pressure for a period of 5 to 10 minutes without rubbing the surface in a manner that would disturb any chalking. The test panels shall the be dried for 24 hours at ambient laboratory conditions. The following overcoats shall be brush applied to 1/2 of each of two panels and the overcoated panels shall be cured for 48 hours at ambient laboratory conditions:

(a). One coat of enamel, in conformance with this specification, at a nominal wet film thickness of 3 to 4 mils (0.003 to 0.004 in).

(b). One coat of MIL-E-24635 Haze Gray no. 27 (FED-STD-595 Color no. 26270) at a nominal wet film thickness of 3 to 4 mils (0.003 to 0.004 in).

The adhesion between test coats shall be determined in accordance with the method described in 4.6.11.1. The adhesion and appearance shall conform to the requirements of 3.8.10.

4.6.13 Accelerated weathering. The test panels shall consist of four panels prepared in accordance with 4.5.1.1. The priming and coating with enamel shall be in accordance with 4.6.11.1. Measure the initial color of the test panel in accordance with ASTM E 97 on any suitable instrument. The panels shall then be exposed in an apparatus in conformance with ASTM G 53 for a total of 300 hours. The cycle shall be 4 hours ultra-violet exposure using UVA-313 bulbs at 60°C (140°F) and 4 hours water condensation only at 40°C (104°F). Remeasure color and gloss of the tested panels. Calculate change in gloss. Calculate color differences in accordance with ASTM D 2244. Results shall comply with the requirements of 3.7.2.

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4.6.14 Hazardous material determination. Soluble and total metal content shall be determined on a dried powderized enamel in accordance with the requirements of 4.6.14.1 and 4.6.14.2. The hazardous solvent content shall be determined in accordance with the requirements of 4.6.14.3.

4.6.14.1 Toxicity characteristic leaching procedure. Soluble metal content shall be determined on a dried powderized enamel in accordance with the 40 Code of Federal Regulation (CFR), CH 1, Part 261, Appendix II Method 1311, toxicity characteristic leaching procedure (TCLP), and the soluble metal content shall not exceed values which would cause the material to be classified as a hazardous waste.

4.6.14.2 Waste extraction test. Soluble and total metal content shall be determined on a dried powderized enamel in accordance with California Administrative Code, Title 22, the waste extraction test, the soluble metal content, and the total metal content shall not exceed values which would cause the material to be classified as a hazardous waste (see tables I and II).

4.6.14.3 Hazardous air pollutants. Hazardous solvent content shall be determined on the extracted liquid of the enamel in accordance with the 40 Code of Federal Regulation (CFR), CH 1, Part 261, Appendix II Method 1311, toxicity characteristic leaching procedure (TCLP). The test results for each solvent shall be in accordance with the requirement in 3.4.3.

4.6.15 Long term exterior exposure. Test panels shall be of aluminum having nominal dimensions of 150x300x3 mm (6x12x1/8 in). Test requires 4 primed test panels. Test panels shall be degreased in accordance with methods B, C or D of ASTM D 609. The degreased test panels shall be abrasive blasted with new, clean aluminum oxide to provide a profile pattern of 0.025 to 0.076 mm (0.001 to 0.002 in) (average). All test panels shall be primed with 2 spray applied coats of MIL-P-24441/1 Type III or IV F.150 Green primer. Each coat of MIL-P-24441/1 Type III or IV F.150 shall be applied to 0.051 to 0.076 mm (0.002 to 0.004 in) dry film thickness. Two coats of the test enamel shall be spray applied on the MIL-P-24441/1 Type III or IV F.150 coated panels to obtain a final dry film thickness of the test topcoat of 0.101 to 0.152 mm (0.004 to 0.006 in). The test panels shall be cured for not less than 24 hours between application of the coats of test topcoats at ambient laboratory conditions. Spraying properties shall be in accordance with 3.8.5, as appropriate. The completed painted test panels shall be cured for at least 7 days prior to mounting on the test rack. Test specimens shall be mounted on a test fence having a 45° south facing rack at a distance of not greater than 100 feet of an ocean high tide line. The exposure shall be in accordance with paragraph 6 of ASTM D 1014. The fence shall be at the Naval Research Laboratory, Marine Corrosion Laboratory, Key West, Florida, USA. Prior to initiating the test, photograph all of the test panels (include a gray color scale in all photos and develop photographs to the scale). After a test period of one year (365 days), examine for chalking in accordance with ASTM D 4214, Fig 2 after the panels are rinsed with cold tap water at nominal tap pressure for a period of 5 to 10 minutes without rubbing the surface in a manner that would disturb any chalking. Then wash the panels with a warm soap solution using a soft sponge to remove any chalking or dirt. The test panels shall the be dried for 24 hours at ambient laboratory conditions and examine for film deterioration and color as follows:

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PROPERTY	ASTM TEST METHOD
Color	E 97
Gloss - 60°	D 523
Checking	D 660
Cracking	D 661
Blistering	D 714
Color difference	D 2244.

The test results shall conform to the requirements of 3.7.4.

4.6.16 Impact resistance The impact resistance of the enamel shall be determined in accordance with ASTM D2794. A suitable instrument is the Gardco Model 172 Universal Impact tester with Model 172 -GE Impact tester Adapter complete with indenter plate, fixture plate, and rubber anvil pad. The GE impacter weighs 3.6 pounds and has formed on each end four convex spherical segments which are of different radii and extensions. Test panels shall be in accordance with 4.5.1.1. With an uncoated panel in place on the rubber anvil pad, determine the height of drop for each end of the impacter to show on the panel slight evenly spaced segment marks of the impacter circumference. The heights so determined will always be used. Place the coated panel, film downward, on the rubber anvil pad at the bottom of the impacter guide. Drop the impacter on the panel from the above previously determined height. Drop the impacter end and drop it on the panel adjacent to the first area of impact. Remove the panel form the tester and inspect the coating. Use 10 power magnification to detect fine surface cracking. Record the maximum percent increase corresponding to the largest spherical impression at which no cracking occurs or other failure.

4.6.17 Compatibility test. The thoroughly mixed enamel shall be allowed to stand undisturbed for 5 minutes. On one clear plate-glass panel, prepared in accordance with method 2021 of Fed. Test Method Std. No. 141, brush a coat of the mixture to approximately 25 um (0.001 in) dry film thickness and allow to dry at room temperature in a vertical position for 24 hours. While brushing, observe for streaks and pigment separation. On another panel prepared in the same way, draw down a 51 um (0.002 in) wet film thickness on a glass panel. Allow to stand five minutes in a horizontal position. While the paint is still wet, rub-up an area using the index finger in a circular motion for a minimum of 20 revolutions. Exert light pressure of the finger while rubbing so as not rub off the film. Allow the enamel to dry for 24 hours and compare the rubbed-up area against the unrubbed-up area. Any visual difference in color or gloss, streaks, craters, or pigment floating of the dried film shall be cause for rejection due to incompatibility. The results of the compatibility tests shall be evaluated for conformance in accordance with the requirements of 3.8.13.

4.6.18 60-degree specular gloss. The thoroughly mixed enamel shall be drawn down on plane, opaque, white glass panels as specified in method 2021 of FTMS-141. Use a film applicator that will produce a nominal wet film thickness of 0.075 mm (0.003 inch). Dry for a minimum of 24 hours under referee conditions in a dust free environment. Paint specimen shall be dry hard before gloss test. Use longer cure time if required. When tested as specified in table V, gloss shall conform to requirements of table IV as applicable, for class specified.

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4.6.19 Quantitative tests. The enamel shall be tested for the quantitative requirements of table V as specified in 3.7 as follows (see table IV):

Characteristic	ASTM/Test method
Flash point	D 3278
Course particles	Method 4092.1 of FTMS-141
Consistency	D 562
Fineness of grind	D 1210
Contrast ratio	D 2805

The measured characteristic values shall be in accordance with the requirements of table V.

4.7 Toxicological assessment of product. To determine conformance to requirements of 3.4.1 and 3.4.2, the manufacturer of the paint shall disclose the formulation of his product to the Navy Environmental Health Center, Code 34, 2510 Walmer Avenue, Norfolk, VA 23513-2617. The disclosure of proprietary information, which shall be held in confidence shall include: the name, formula, CAS number, and approximate percentage by weight and volume of each ingredient in the product (totalling 100 percent); the results of any toxicological testing of the product; identification of its pyrolysis products; and any other information as may be needed to permit an accurate appraisal of any toxicity problem or issues associated with the handling, storage, application, use, removal, disposal, or combustion of the material. Where possible, Chemical Abstract Service (CAS) numbers should be provided for all ingredients in the formulation. A current MSDS and Hazardous Chemical Warning Label, conforming to the provisions of the OSHA Hazard Communication Standard, shall be provided for each product. In addition, the manufacturer shall provide a current MSDS for each ingredient used in the formulation. Within 30 days prior to data submission, the manufacturer shall contact the supplier of each ingredient, and confirm each MSDS submitted is current. a statement documenting this contact shall be affixed to each ingredient MSDS, and indicate: (1) the date of the contact and (2) the name of the company representative, with telephone number, who confirmed the MSDS was current. The OSHA Form 20 is obsolete, therefore all MSDS must be submitted on an OSHA Form 174, or an equivalent form containing the identical data elements. Where available, copies of industrial hygiene reports, which address potential health hazards related to working with the material, are also to be submitted as part of the documentation package. Information pertaining to any adverse health effects experienced by company workers during research/development and manufacture of the product, and industrial experience with the product by major commercial users, is also requested. All data provided must be clearly identified to show it is being provided in connection with qualification under MIL-E-24763A.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions.)

5.1 Packaging.

The enamel shall be package level A, B, or C packed as specified (see 6.8) and marked in accordance with PPP-P-1892, herein, and shall include bar codes, hazardous chemical warning label (see 3.4) and applicable packaging acquisition options therein specified (see 6.2). The enamel shall be furnished in metric or English containers equivalent to 1-quart or 1-gallon, multiple friction top, resin-lined containers, or in 5-gallon, lug cover, steel, resin-lined pails, or 55-gallon, resin-lined drums, as specified (see 6.2).

In addition, for Navy acquisitions, the following Navy fire-retardant requirements apply:

(a) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood, including laminated veneer materials, used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material in accordance with MIL-L-19140 as follows:

Levels A and B: MIL-L-19140, type II (weather resistant), Category I (general use).

Level C: MIL-L-19140, type I (non-weather resistant), Category I (general use).

(b) Fiberboard. Fiberboard used in the construction of interior (unit and intermediate) and exterior containers, including interior packaging forms, shall conform to the PPP-F-320. PPP-F-320 classes shall be domestic fire-retardant or weather resistant fire-retardant as specified (see 6.2).

5.1.1 Commercial. The enamel shall be furnished in metric or English equivalents of 1-quart or 1-gallon, multiple friction top, resin-lined container or in 5-gallon, lug cover steel, resin-lined pails, or 55-gallon, resin-lined drums, as specified (see 6.2). If specifically allowed by the contracting activity, commercial plastic containers and pails may be utilized.

5.2 Marking. Marking shall be at level A or B in accordance with MIL-STD-129 (see 6.2).

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5.2.1 Preliminary markings. In addition to the markings required in accordance with MIL-STD-129, FED-STD-313, and 29 CFR Parts 1910.1200, 1915.99, 1917.28, 1918.90, and 1926.59, all individual containers shall have the following markings (see 6.2):

"CAUTION: This enamel may contain certain volatile solvents with possibly hazardous vapors. Use with adequate ventilation. Avoid prolonged breathing of vapors or spray mists. Consult MSDS and Product label for specific information regarding potential health hazards related to working with this material, and precautions which must be taken for the material to be used safely"

5.2.2 Special marking. In addition to the markings required by the contract or order (see 6.2), each container, interior and exterior shall be marked with the following (see 6.2):

5.2.2.1 User instruction marking. In addition to the markings specified in section 5.2, all primary containers shall be legibly labeled with the manufacturer's mixing, and application instructions. Thinning of the product shall not be necessary. Application instructions must include the following points and be consistent with the required ASTM F 718 data sheet (see 3.3):

(a) Requirement that the surface be primed, prior to application of the enamel and that it be free of millscale, corrosion, loose paint, dirt, oil, grease, salt deposits and moisture.
(b) Indicate the minimum and maximum surface temperatures for application and required dew point levels (if any).
(c) Requirement, if applicable, for equipment to be adequately grounded and spray equipment cleaning instructions.
(d) Disposal of debris and container.

5.2.2.2 Type I special marking. Type I enamels shall be marked in bold red lettering as follows:

"Contains a maximum of **INSERT PROPER NUMBER HERE "NOT GREATER THAN 340"** grams per liter of volatile organic content per 40 CFR CH. 1, Part 60, Appendix A, (U.S. EPA) Method 24."

5.2.2.3 Type II special marking. Type II enamels shall be marked in bold light blue lettering as follows:

"Contains a maximum of **INSERT PROPER NUMBER HERE "NOT GREATER THAN 250"** grams per liter of volatile organic content per 40 CFR CH. 1, Part 60, Appendix A, (U.S. EPA) Method 24."

5.2.3 Data sheet. A copy of the material safety data sheet (MSDS), Navy Environmental Health Center findings and the ASTM F 718 data sheet shall be attached to the shipping document for each destination (see 3.2 and 3.3).

5.2.4 Color of labels. The major bold lettering on all labels shall be brown on white background for class 1, orange on white background for class 2, green on white background for class 3, and blue on white background for class 4.

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5.2.5 Precautionary marking. Unit, intermediate and shipping containers shall be marked with the applicable precautionary information detailed in the American National Standard ANSI Z 129.1. Shipping containers shall be marked in accordance with Department of Transportation regulations as specified in 49 CFR 171-178 and shall bear the "Flammable Liquid" red label when necessary as specified therein. All unit and intermediate packs of toxic and hazardous chemicals and materials shall also be labeled in accordance with the applicable laws, statutes, regulations or ordinances, including Federal, State, and municipal requirements. In addition, unit and intermediate containers, including unit containers that serve as shipping containers such as pails and drums, shall be marked exactly like the general example detailed in American National Standard ANSI Z 129.1.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers a high grade, acrylic, water bound, emulsion type (latex) enamel. Type I is intended for use where the VOC shall not greater than 340 g/L and type II is intended for use where the VOC shall not greater than 250 g/L (see 1.2). Paints to this specification are intended for use over primed metal, particularly on smooth, exterior metal, such as ship exterior topside metalwork and equipments. It is intended to be highly weather resistant, and characterized by excellent color and gloss retention, good drying, freedom from after tack, and good flexibility. The enamel may be applied with brush, roller or spray. This enamel is not intended for painting ship interior bulkheads or overheads; it is not fire retardant.

6.1.1 Additional intended uses. Some additional intended uses of the enamel meeting this specification are for machinery, refinishing trucks and buses, passenger and freight cars, metal drums, metal signs, metal railing and fences, marine use above water, metal trim (exterior), metal sidings, metal doors and bucks, metal structures (exterior), and properly primed wood (exterior).

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Color desired (see 1.1, 3.7.1 and table V).
(c) Type and class of enamel required (see 1.2).
(d) Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(e) QPL number and requirement for qualification (see 3.1).
(f) Level of packaging, level of packing and packaging acquisition option(s) required (see 5.1).
(g) When fire retardant treated lumber and plywood is not required (see 5.1a).
(h) Class of fire retardant fiberboard required (see 5.1b).
(i) Size of container required (see 5.1.1).
(j) Required marking (see 5.2, 5.2.1 and 5.2.2).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.3., 3.4, 4.4 and appendix	DI-MISC-80678	Certification/ data report	10.3.1 does not apply
4.3	DI-NDTI-80809	Test/Inspection reports	----------

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-24763 whether or not such products have actually been listed by that date. The attention of contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 03Q22, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4.1)

6.4.1 Provisions governing qualifications. Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Standardization Document Order Desk.

6.5 Color numbers and descriptions. Color number and descriptions are as shown in table VI.

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TABLE VI. FED-STD 595 Color numbers and descriptions. 1/ 2/3/

Gloss	Semigloss	Low gloss lusterless	Word Description
---	*20109	30109	Brown (20109 for replacement of DOD-E-18210 formula 23)
---	20117	30117	Brown
10080	---	---	Brown (piping and valve marking)
10324	---	---	Tan (piping and valve marking)
10371	---	---	Spar (U.S. Coast Guard) or Buff (piping and valve marking)
11105	---	---	Red (piping and valve marking)
11120	---	---	Red (OSHA safety color)
11136	21136	31136	Red (Insignia red)
12197	---	---	International orange
12199	---	---	Red (U.S. Coast Guard)
12246	---	---	Orange (piping and valve marking)
12300	---	---	Orange (OSHA safety color)
13538	---	---	Yellow (piping and valve marking)
13655	23655	33655	Yellow
13591	---	---	Yellow (OSHA safety color)
14062	---	---	Dark green (piping and valve marking)
14097	24097	*34097	Dark green (34097 for replacement of DOD-E-18214 Formula 19
14120	---	---	Green (OSHA safety color)
14449	---	---	Light Green (piping and valve marking)
15042	25042	35042	Blue (Sea blue)
15044	---	---	Dark blue (piping and valve marking)
15048	25048	35048	Blue (Insignia blue)
15092	---	---	Blue (OSHA safety color)
15102	25102	---	Blue (Light blue)
15182	---	---	Blue (U.S. Coast Guard)
15200	---	---	Light blue (piping and valve marking)
---	*26008	---	Gray (26008 for replacement of DOD-E-699 formula 20)
16081	---	---	Dark gray (piping and valve marking)
16099	---	---	Deck gray (U.S. Coast Guard)

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---	26118	*36118	Gray # 11 (36118 for replacement of DOD-P-15183 formula 109)
---	*26173	36173	Gray # 17 (Ocean gray)
16187	---	---	Gray (piping and valve marking)
---	26231	*36231	Gray # 23 (36231 for replacement of DOD-E-700 formula 20L)
---	---	36270	Lusterless Gray # 27 (Haze gray)
16307	*26307	36307	Gray # 30 (Bulkhead gray)
---	*26373	36273	Gray # 37
16376	---	---	Light gray (piping and valve marking)
16492	*26492	36492	Gray # 49 (Gull or Pearl Gray)
17038	*27038	37038	Black (27038 for replacement of DOD-E-698 formula 24; 37038 for replacement of DOD-P-15146 formula 104; 17038-U.S. Coast Guard)
17043	---	---	Gold (piping and valve marking)
17100	---	---	Dark purple (piping and valve marking)
17142	---	---	Light Purple (piping and valve marking)
17155	---	---	Purple (OSHA safety color)
17875	*27875	37875	White (Insignia white - 27875 matches color of DOD-E-1115 formula 30)
17886	27886	37886	White (bone white)
23814	----	----	Chartreuse (piping and valve marking)

1/ See FED-STD-595 for color cards.

2/ * mark Navy colors.

3/ Table VI provides FED-STD-595 color numbers and descriptions for commonly specified colors. To order colors not listed, Contracting officer shall provide FED-STD-595 color number or other information necessary to identify the product desired.

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6.6 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 Occupational reproductive hazards. A copy of the OPNAVINST 5100.23C-change 1 to chapter 28, occupational reproductive hazards can be obtained from Office of the Chief of Naval Operations, Code N454D, Washington, DC 20350-2000.

6.8 Conditions for use of level B preservation. When level B preservation is specified (see 5.1), this level of protection should be reserved for the acquisition of enamel for resupply worldwide under known favorable handling, transportation, and storage conditions.

6.9 Shelf life For the cataloging, logistics, and storage functions of Federal acquisition and supply agencies, materials acquired under this specification shall be assigned a 1-year non-extendable shelf life (shelf life code H) beginning at the date of manufacture.

6.10 Volatile content. Although the container marking specifically refers to Federal regulations, the paint may be used any where else a product complying with 1.2 is allowed. This includes other air pollution control districts or similar areas controlling the emission of solvents volume into the atmosphere under volatile organic content (VOC) regulations.

6.11 Subject term (key word) listing.

Acrylic
Flat gloss
Gloss
Latex
Low gloss
Paint
Semigloss
Top coat
VOC

Preparing activity:
Navy - SH
(Project 8010-N425)

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APPENDIX A

CERTIFICATION/DATA REPORT TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the content requirements of the certification to accompany each lot of enamel submitted for Government acceptance. This appendix is mandatory only when data item description DI-MISC-80678 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. TECHNICAL CONTENT REQUIREMENTS

30.1 Certificate of compliance. The certificate of compliance shall include separate and specific statements by the contractor that:

(a). The VOC conforms to the requirements of the enamel type specified in table V.

(b). The Lead, Chromium, Chromium (III), Chromium (VI) and Asbestos content conforms to the requirements of the enamel specified in tables II,III and V.

(c). The hazardous air pollutant content conforms to the requirements for the enamel specified in table I.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the comment number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of this form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-E-24763A(SH)	2. DOCUMENT DATE (YYMMDD) 93/08/25

3. DOCUMENT TITLE
ENAMEL, EMULSION TYPE, FOR SHIPBOARD USE

4. NATURE OF CHANGE (identity paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC) DALE SOWELL, SEA 03M ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: DSN: TPOC: 703-602-0214 8-332-0214
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND ATTN: SEA 03Q42 2531 JEFFERSON DAVIS HIGHWAY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone 703-756-2340 DSN 289-2340

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